Nisong Monyimba

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github.com/NisongMonyimba/AppliedMathThesis

Applied mathematician with a background in biomedical engineering whose research focuses on stochastic analysis, mean-field games, stochastic control, and scientific computing. My current work develops analytical and computational methods for common-noise mean-field games, combining stochastic differential equations, optimal transport, numerical analysis, and reproducible research software. Developing an independent research program in stochastic analysis, stochastic control, and scientific computing, with the goal of an academic career in applied mathematics.

Research Interests

Stochastic Analysis • Mean-Field Games • Numerical Analysis • Scientific Computing • Stochastic Control • Optimal Transport • Partial Differential Equations

Education

Biomedical Engineering, M.Sc. — Arizona State University Graduated May 2022

Ira A. Fulton Schools of Engineering; MasterCard Foundation Scholar; Dean's List: Fall 2020, Spring 2021

Biomedical Engineering, B.Sc. — KNUST Graduated May 2021

Kwame Nkrumah University of Science & Technology; MasterCard Foundation Scholar; First Class Honours

Undergraduate quantitative coursework: Calculus, Differential Equations, Numerical Methods, Probability and Statistics, Signals and Systems.

Independent Mathematical Preparation

Graduate-Level Self-Study 2025–Present

Undertaken independently of formal coursework, in connection with the research program below

- Real Analysis, Functional Analysis, Measure-Theoretic Probability, Stochastic Calculus, Mean-Field Games, McKean–Vlasov SDEs, Optimal Transport, and Numerical PDEs

Research

Independent Research Program in Applied Mathematics 2025–Present

Mean-field games • stochastic control • Wasserstein geometry • optimal transport • FBSDEs • numerical PDEs • Lean 4 formal verification

Current research develops mathematical and computational methods for common-noise mean-field games, combining stochastic analysis, numerical analysis, scientific computing, and reproducible research software. Current outputs: a theoretical preprint, a numerical preprint, a software repository, and an independent research monograph (below).

Publications and Preprints

Monyimba, N. (2026). *Quantitative Contraction Estimates for Common-Noise Mean-Field Games.* Preprint (to be posted). Establishes a local contraction estimate, on an explicit time horizon, for the fixed-point map characterising equilibria of mean-field games with common noise, extended to arbitrary horizons via a continuation argument.
PDF | GitHub | DOI: 10.5281/zenodo.21131132

Monyimba, N. (2026). *Numerical Approximation of Common-Noise Mean-Field Games via Particle Methods and Finite Differences.* Preprint (to be posted). Combined Euler–Maruyama, particle-method, and finite-difference scheme for common-noise mean-field game equilibria, with every reported convergence rate the direct, reproducible output of executed code.
PDF | GitHub | DOI: 10.5281/zenodo.21131132

Independent Research Monograph

Quantitative Convergence Analysis of Stochastic Maximum Principles for Mean-Field Games with Common Noise (2026). A book-length treatment developed independently following the 2022 M.Sc., covering well-posedness, particle methods, finite-difference schemes, and partial Lean 4 formalisation.
PDF | Full repository

Current Research Directions

- Nonlinear (non-LQ) common-noise mean-field games
- Numerical stochastic control and common-noise master equations
- Numerical methods for high-dimensional FBSDEs and PDEs
- Formal verification of stochastic analysis results in Lean 4

Research Software

AppliedMathThesis (Python, Lean 4)

Research software implementing numerical methods for common-noise mean-field games

- SDE solvers (Euler–Maruyama, Milstein), including mean-field/common-noise variants
- Particle methods with conditional propagation-of-chaos verification
- Finite-difference Fokker–Planck solver
- Wasserstein-distance convergence verification and automated numerical benchmarks
- Reproducible computational experiments; version-controlled, continuously integrated (GitHub Actions), automatically tested (pytest)
- Lean 4 formalisation of core estimates (Gronwall, contraction bound, propagation of chaos)
- github.com/NisongMonyimba/AppliedMathThesis | DOI: 10.5281/zenodo.21131132

Research Experience

Research Assistant, Weaver Lab, Arizona State University 2021–2022

- Analyzed trophoblast proliferation dynamics by quantifying cell counts over time and fitting growth curves.
- Used Fiji to extract quantitative metrics from laboratory images, including colony expansion area and fluorescence intensity, supporting rigorous modeling of cellular growth patterns.
- Quantified cell proliferation rates from fluorescence microscopy data using Fiji and GraphPad Prism; developed statistical plots and regression analyses to characterize proliferation and viability.
- Developed Python-based probabilistic simulations and Monte Carlo analyses to evaluate variability in material properties and stochastic loading, improving predictive accuracy.

Research Assistant, Dr. Pizziconi's Lab, Arizona State University 2022–2023

- Designed and optimized bone fixation screws using Halpin–Tsai micromechanics models to predict composite behavior under mechanical load.
- Created and refined 3D CAD models in SolidWorks, ensuring dimensional accuracy, manufacturability, and integration with experimental validation workflows.
- Trained and mentored lab members in modeling, simulation, and experimental validation, connecting theoretical mathematics with engineering practice.
- Collaborated with interdisciplinary teams to iterate designs from simulation results; documented and presented research outcomes supporting publication efforts and potential clinical translation.

Research Assistant, Medical Device Regulation (MDR) Project, KNUST 2020–2021

- Analyzed regulatory frameworks governing medical devices across 20 African countries, identifying disparities in approval processes, safety standards, and market accessibility.
- Used publicly available regulatory data to categorize countries by regulatory maturity, from fully established authorities to emerging or informal frameworks.

Med Tech R&D Engineer, Mayo Clinic & Arizona State University 2022–2023

- Designed bone fixation screws using Halpin–Tsai equations and Mori–Tanaka models.
- Ran nonlinear FEA simulations to analyze fatigue, stress distribution, and deformation behavior.

Teaching Philosophy

I aim to teach mathematics by combining rigorous theory with computational examples and reproducible scientific programming.

Teaching Experience

Teaching Interests

Based on completed undergraduate coursework, self-directed graduate-level study, and the teaching and tutoring experience below

- Calculus I & II, Linear Algebra, Differential Equations, Numerical Analysis, Probability & Statistics, Engineering Mathematics, Scientific Computing (Python)

Teaching Assistant, KNUST 2019

- Tutored biomedical engineering undergraduates in calculus, electrical circuits, and biostatistics; assisted with lab supervision, grading, and practical demonstrations.

Science & Math Teacher, NSMQ Team, Kwanyako Senior High School 2016–2018

- Prepared and led students for regional and national science and math competitions, emphasizing teamwork, critical thinking, and problem-solving.

Math Teacher, Witness Academy 2016–2017

- Taught core mathematics and physics; prepared junior-high students for final exams (BECE).

Master's Applied Project & Undergraduate Capstone Project

Master's Applied Project: Trophoblast-on-Chip Microfluidic Device 2022

- Investigated trophoblast growth to quantify proliferation dynamics (cell counts over time, growth curves, proliferation index) using Fiji to extract quantitative growth metrics from laboratory cell images.
- Performed statistical analysis including growth-curve comparison across conditions (two-way ANOVA), doubling-time comparison between treatments (unpaired t-test), and proliferation-marker quantification (Kruskal–Wallis).
- Developed microfluidic channel designs in SolidWorks, simulated laminar flow and shear profiles in COMSOL Multiphysics, and validated models against analytical equations.

Capstone Project: Novel ECMO Cannula Design, Arizona State University 2021

- Designed and simulated ECMO cannula flow in COMSOL Multiphysics, optimizing shear stress and velocity distribution; built a 3D model in SolidWorks and simulated flow in ANSYS Fluent.
- Quantified flow uniformity and reduced clot-formation risk by optimizing inlet geometry and flow-rate parameters.
- Performed finite element analyses (FEA) in ANSYS and SolidWorks to assess performance and safety margins under physiological load conditions.

Internship Experience

REU Internship, Case Western Reserve University 2020

- Studied biological design and product development principles; designed Arduino-based air-quality sensors and prototyped PCB systems using EAGLE CAD.
- Collaborated with interdisciplinary teams to apply quantitative modeling to life-science problems.

BME Internship, Komfo Anokye Teaching Hospital, Kumasi, Ghana 2019

- Supported medical equipment maintenance and calibration; performed voltage and current testing.
- Documented device performance data and proposed engineering solutions to minimize error variance; presented findings to senior biomedical engineers and clinicians.

Engineering Internship, Vodafone (Telecom) Headquarters, Accra, Ghana

- Designed and proposed a telemedicine smartphone application concept for patient monitoring.
- Assisted in vendor management and quality audits for customer-facing telecom hardware systems.

Professional Experience

Senior Manufacturing Engineer, Abbott Laboratories 2024–2025

- Optimized medical device grinding precision by 31.7% through DOE, ANOVA, and SPC-driven process improvements.
- Applied Monte Carlo simulations, PCA, and variance component analysis to identify and mitigate process variability.

Quality Engineer, Vastek Inc., San Diego, CA 2023–2024

- Designed devices and implemented engineering design practices targeting efficiency improvements; worked with a cross-functional team of medical professionals and engineers.
- Conducted FMEA, Bayesian risk analysis, and fault-tree analysis to quantify cascading failure probabilities.

Volunteering, Community Service & Projects

- **Group Homes Volunteer (Autism Support):** assisted residents with learning and recreational activities.
- **MasterCard Foundation Community Projects:** led sanitation, clean-water, and youth education projects.
- **Oforikrom Outreach Project:** organized public school clean-ups, donations, and science tutoring.
- **Vocational Training:** trained four youths in electrical wiring and biofill installation.
- **Personal Tutoring Initiative:** mentored underprivileged students in mathematics; three students gained high-school admission and two gained college admission, all first-generation in their families.
- Organized community cleaning (gutters, weeds, trash) in Ashaiman-Zenu to reduce contamination and malaria risk.

Awards & Fellowships

- MasterCard Foundation Scholarship, Arizona State University and KNUST
- MORE Fellowship Award, Arizona State University
- Vodafone Telecommunications Design Competition, 1st Place
- 2nd Place, KNUST Engineering & Medical School Innovation Challenge
- Dean's List, Arizona State University

Leadership & Mentorship

- Peer Mentor, MasterCard Foundation Scholars (KNUST & ASU)
- Community Project Facilitator, MasterCard Foundation Ghana

- Member, Young African Leaders Initiative (YALI)
- Club Executive, Biomedical Engineering Society (KNUST & ASU)
- Leader, NSMQ Science & Math Quiz Team, Kwanyako SHS

Certifications

- Lean Six Sigma (Green Belt)
- Design of Experiments (Minitab & JMP)
- Global Leadership & Project Management (YALI / ASU Global Leadership Academy)

Professional Memberships

Society of Manufacturing Engineers (SME) • National Society of Black Engineers (NSBE) • Biomedical Engineering Society (BMES) • ASU Global Leadership Academy • Toastmasters International • KNUST Peer Counsellors & MasterCard Peer Mentors • Ghana Engineering Students Association • TraTech

Mathematical Computing

Python (NumPy, SciPy, pandas, statsmodels) • Lean 4 • MATLAB • GNU Octave

Scientific Software

Git/GitHub • GitHub Actions (CI) • pytest • LaTeX (Overleaf) • R

Scientific Computing

Finite-Difference Methods • Monte Carlo Simulation • Euler–Maruyama & Milstein Schemes • Particle Methods • Numerical Linear Algebra • Computational Complexity Analysis • Numerical Verification • Reproducible Research

Formal Mathematics

Lean 4 • formalisation of selected results in stochastic analysis, including Gronwall's inequality, contraction bounds, and propagation-of-chaos lemmas. Repository includes an automated compilation check (GitHub Actions, `lake build`) against current Mathlib releases.

Engineering Software

SolidWorks • ANSYS Fluent • COMSOL Multiphysics • Autodesk Fusion 360 • SimScale • finite element methods

Technical Skills

Statistical & Process Methods: ANOVA, PCA, DOE, SPC, optimization, GraphPad Prism, Minitab, JMP.

Other: Fiji/ImageJ, BioRender, Figma, Inkscape, LabVIEW, LTSpice, Windchill, Zotero.

Languages

English • Twi